

IT615 – Data Base Management System
Dhirubhai Ambani Institute of Information and Communication Technology (DA-IICT)
Second In-Sem Examination, October 2023

[Time – 2 Hours]

[Total Marks - 60]

Instructions:

- Answer all question. All questions are self-explanatory and understanding of question is a part of evaluation.
- No query regarding questions will be entertained during examination by course instructor or invigilator.

1. Explain how Functional dependencies can be used to indicate the following. Explanations should be done with proper example. [Attribute within entity set can be assumed by yourself, and must be mentioned in answer sheet]

- (A). A one-to-one relationship set exists between entity sets “*student*” and “*instructor*”.
(B). A many-to-one relationship set exists between entity sets “*student*” and “*instructor*”

[5+5 = 10 Marks]

2. Consider the following set F of functional dependencies on the relation schema (A,B,C,D,E,G)

$A \rightarrow BCD$

$BC \rightarrow DE$

$B \rightarrow D$

$D \rightarrow A$

- (A). Compute F^+ .
(B). Compute B^+ .
(C). Prove (using Armstrong’s axioms) that AG is a superkey.
(D). Compute a canonical cover for this set of functional dependencies F; give each step of your derivation with an explanation
(E). Give a 3NF decomposition of the given schema based on a canonical cover.
(F). Give a BCNF decomposition of the given schema using the original set F of functional dependencies.

[4 + 4 + 4 + 8 + 5 + 5 = 30 Marks]

3. Let $ri(z)$ and $wi(z)$ denote read and write operations respectively on a data item z by a transaction T_i . Consider the following two schedules.

S1 : $r1(x) r1(y) r2(x) r2(y) w2(y) w1(x)$

S2 : $r1(x) r2(x) r2(y) w2(y) r1(y) w1(x)$

In these two different schedules, which is conflict serializable, and which is view serializable. Give answer with proper justification (Without Justification answer will treated as wrong)

[4 Marks]

4. Consider the relation $R(P, Q, S, T, X, Y, Z, W)$ with the following functional dependencies.

$PQ \rightarrow X$

$P \rightarrow YX$

$Q \rightarrow Y$

$Y \rightarrow ZW$

Consider the decomposition of the relation R into the constituent relations according to the following two decomposition schemes.

$D1 : R = [(P, Q, S, T); (P, T, X); (Q, Y); (Y, Z, W)]$

$D2 : R = [(P, Q, S); (T, X); (Q, Y); (Y, Z, W)]$

In these decomposition, which decomposition is Lossy decomposition and which is lossless decomposition.

Give answer with proper justification (Without Justification answer will be treated as wrong)

[5 Marks]

5. Consider the transactions $T1, T2$, and $T3$ and the schedules $S1$ and $S2$ given below.

$T1: r1(X); r1(Z); w1(X); w1(Z)$

$T2: r2(Y); r2(Z); w2(Z)$

$T3: r3(Y); r3(X); w3(Y)$

$S1: r1(X); r3(Y); r3(X); r2(Y); r2(Z); w3(Y); w2(Z); r1(Z); w1(X); w1(Z)$

$S2: r1(X); r3(Y); r2(Y); r3(X); r1(Z); r2(Z); w3(Y); w1(X); w2(Z); w1(Z)$

In these two different schedules, which is conflict serializable. **Give answer with proper justification (Without Justification answer will be treated as wrong)**

[5 Marks]

6. Which of the following schedules is serializable and recoverable, but does not avoid cascading rollback?

(A) $r_1(X), r_2(X), w_1(Y), r_2(Y), r_1(Z), w_2(X), c_2, w_1(Z), c_1$

(B) $r_1(X), r_2(X), w_2(Y), r_2(Z), r_1(Y), w_1(X), c_1, w_2(Z), c_2$

(C) $r_2(X), r_1(X), w_1(Z), r_2(Z), r_1(Y), w_1(Y), c_1, w_2(X), c_2$

(D) $r_1(X), r_2(X), w_1(Y), w_1(X), w_1(Z), c_1, r_2(Y), w_2(Z), c_2$

Give answer with proper justification (Without Justification answer will be treated as wrong)

[4 Marks]

7. File has $r = 20,000$ STUDENT records of *fixed length*. Each record has the following fields: Name (30 bytes), Ssn (9 bytes), Address (40 bytes), PHONE (10 bytes), Birth_date (8 bytes), Sex (1 byte), Major_dept_code (4 bytes), Minor_dept_code (4 bytes), Class_code (4 bytes, integer), and Degree_program (3 bytes). An additional byte is used as a deletion marker. Calculate the record size, and file size in bytes.

[2 Marks]